Response to Reviews

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The Occurrence Birth-Death Diffusion Process:
Unraveling Diversification Histories with Fossils and Heterogeneous Rates

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Dear Editor, Associate Editor, and Reviewers,

Thank you for the thoughtful and constructive reviews of our manuscript. We are grateful for the opportunity to revise and resubmit. We have carefully addressed all comments below, and all changes in the revised manuscript are highlighted in blue. Our responses appear in blue below.

Associate Editor (Dr. April Wright)

> *Like Dr. Wagner, I was not enamored of the terms "phylogenetic fossil" and "occurrence fossil." With occurrence fossils affecting the probability of trees and tree shapes under the FBD, all fossils are to some degree phylogenetic fossils. I like the terminology of uncoded (the specimens for which there is no coded data) and coded.*

We agree that our previous terminology was confusing and could conflict with the established paleontological meaning of "fossil occurrence." Following the suggestions of the Associate Editor and Reviewer 3, we now use "coded fossils" for those placed in the phylogeny and "uncoded occurrences" for those that are not. This terminology has been updated throughout the manuscript.

> *In Figure 1 of his review, Dr. Wagner points to an issue that I think is the most single important for the authors to consider and address. That issue is that it doesn't seem (in whales) that coded and uncoded fossils actually do come from different distributions and therefore need to be modeled differently. [...] So, in my opinion, there's a very central "Why?" hanging over this manuscript.*

We address this concern in detail in our response to R3.2. In short, our model with separate rates ψ (coded fossils) and ω (uncoded occurrences) is strictly equivalent to a model with a single total fossil sampling rate $\varphi = \psi + \omega$ and a Binomial coding probability $p = \psi/(\psi + \omega)$. The two rates do not imply different sampling distributions or taphonomic regimes; they simply allow the proportion of coded fossils to be estimated from the data rather than fixed in advance. We agree that for cetaceans this proportion likely does not vary much across time periods, and we now clarify this in the revised manuscript (see Methods and Discussion).

> *Both other reviewers point to some unclarity in the description of the augmented tree approach. I think as written, it reads fine, but I agree with Reviewer One that this is an approach that is likely to be unknown to readers, and a figure might be warranted. I'm a moderate on that point – I think it would help readers, but there are a lot of graphics in this manuscript already.*

[TODO]

> *Both reviewers note that Figure 4 makes it seem like the analyses can be run to completion very quickly, which is not the case. I would advise either removing or reconfiguring this figure – this information is communicated more clearly on Table 1. That would free up some space to compare branch rates across analyses, as suggested by Reviewer Two.*

[TODO]

> *One other thing I want to note is that in his review, Dr. Wagner highlights a number of places where this manuscript asserts a sort of naivete about parameters, where long records of literature provide information from which information can be gleaned, or priors derived. For example, using Gamma-distributing sampling rates, when there is a long history in the literature of empirically deriving these, and they tend not to support Gamma-distributions. I think you can make an argument for Gamma and Lognormal being similar enough that it doesn't matter, but the positive argument for doing something that is analytically contraindicated should be made.*

[TODO]

Reviewer 1

R1.1 – Table comparing models

> *It could be helpful to include a table highlighting the difference/similarities between OBDD, OBD, FBDD, and BDD. At the minute the information is sort of scattered across the introduction and methods. A schematic table (features × models) would help readers see exactly what is gained at each step.*

We have added a comparison table (Table 1 in the revised manuscript) at the end of the Introduction, summarizing the key features of all six models (BD, FBD, OBD, BDD, FBDD, OBDD): inclusion of coded fossils, uncoded occurrences, rate heterogeneity (GBM), and piecewise sampling rates. This should help readers quickly see what each model adds.

R1.2 – Augmented tree figure

> *Can the authors maybe spend a bit more time explaining the augmented tree approach, perhaps include a figure. This is a new concept to me and it took a while to understand it. If this is a standard approach you can ignore this, but maybe even a small figure would be really helpful.*

We thank the reviewer for this suggestion. We have added a new figure in the Methods section illustrating the data augmentation approach. The figure shows (A) the empirical tree with coded fossils and uncoded occurrences, (B) one complete data-augmented tree as sampled during an MCMC iteration, with hidden speciation and extinction events and unsampled extant lineages, and (C, D) the latent speciation and extinction rates evolving along the augmented tree.

R1.3 – Computational cost and checkpointing

> *The cetacean case study required extremely long MCMC runs (up to 100 days, Table 1), with some runs being terminated before the finish. This raises some concerns about practical applicability for most users. For example, users with strict walltimes on their clusters. Is checkpointing implemented? Or can you discuss any approaches to increasing computational efficiency that could be implemented down the line?*

[TODO]

R1.4 – Calibration on larger trees

> *The results from the simulation-based calibration study are very impressive but limited to moderate size trees (≤40 tips). Given the cetacean dataset is much larger, would it be worthwhile to include stress tests at higher tree sizes to show that the method remains well-calibrated (or at least does not systematically bias estimates).*

We acknowledge that many of the simulated trees are small, though a substantial number do exceed 40 tips. The primary purpose of the simulation-based calibration is to verify that the core implementation is correct. Once the algorithm is validated, it should scale to arbitrary tree sizes (barring computational limitations). The consistent decrease in estimation error with increasing tree size (Figs. 2 and 3) support this conclusion. We have clarified this point in the text.

R1.5 – Fixed topology stated earlier

> *It is implied in the text but it is not until the discussion that you state that you used a fixed topology, could you say that earlier just for clarity?*

Done. We now state explicitly in the Case Study section that all diversification analyses operate on fixed phylogenies.

R1.6 – Figures 2 and 3 captions

> *Figure 2 and 3 do not have a caption.*

We have expanded the captions of Figures 2 and 3 to describe the calibration study and explain the metrics shown (truth recovery proportions and estimation error).

R1.7 – Typo: "Aquitalian"

> *L 280: "Aquitalian" should be "Aquitanian"*

Corrected, thank you. We also rephrased the epoch names following Reviewer 3's suggestion (see R3.19).

R1.8 – More references on the fossil record

> *In the discussion, especially around the fossil record, I think you could include a few more references.*

[TODO]

Reviewer 2

R2.1 – Figures 2 and 3 captions, NMSE explanation, absolute error

> *Figures 2 and 3: These figures need a more elaborate caption, similar to the other figure captions in the manuscript. Consider reminding the readers that these data are based on the calibration study, and please explain how the normalized mean squared error is calculated, either with an equation or in a sentence. Normalizing and squaring can also hide important behaviour, for example it is possible that extinction and speciation are unbiased, but it is also possible that both are overestimated, or that both are underestimated. You may consider adding another figure to supplementary materials, where you plot the absolute estimation error?*

We have expanded the captions of Figures 2 and 3 to include a description of the calibration study and a definition of the estimation error metric.

[TODO]

R2.2 – Branch-specific rate estimation errors

> *Assuming that one of the main aims of the paper is to estimate branch-specific diversification rates (as in Figure 7), I was surprised that there were no plots of the estimation error in the estimates of the branch-specific rates. [...] I believe the manuscript would be further strengthened if the error in branch-specific rates would also be plotted and discussed.*

The posterior augmented trees from the 500 calibration datasets were not stored, as they are prohibitively voluminous (each tree contains rate trajectories discretised at 10^{-3} Myr along every

branch, including augmented branches). We therefore cannot directly assess per-branch rate estimation errors. We note, however, that under the OBDD model the branch-specific rates are not free parameters: they are determined by the GBM process parameters (λ_0, μ_0, α_λ, α_μ, σ_λ, σ_μ) and by the tree topology. Accurate recovery of the global parameters, as demonstrated in our calibration study (Figs. 2–3), therefore constrains the branch-specific rates as well.

R2.3 – Estimation error vs trend parameter

> *I would be curious to see a plot of estimation errors of branch-specific rates on the y-axis, and the trend parameter on the x-axis. My prediction is that in models where there is a positive trend in net-diversification, the estimation error in branch-specific rates should be smaller [...] However addressing this comment is not essential.*

As noted in R2.2, posterior augmented trees were not stored, so we cannot plot branch-specific rate errors directly. As a proxy, we plotted the NMSE of all global parameters against the true speciation trend α_λ, extinction trend α_μ, and net diversification trend $\alpha_\lambda - \alpha_\mu$ (Supplementary Fig. X). Neither the trend values nor the resulting tree sizes show an obvious influence on estimation error. We include this supplementary analysis for completeness.

R2.4 – Data augmentation: computational advantages and recommendations

> *The authors state that their RevBayes implementation of the OBD model (Andreoletti et al. 2022) is computationally slow [...] Aside from model flexibility – does using augmentation techniques inherently improve the computational efficiency and overall feasibility of the OBD and similar models? Given the authors extensive experience of working with data augmentation techniques, what do you recommend for future developments on diversification inference? Should one use data augmentation or not?*

The clearest advantage of data augmentation is that it simplifies the computation of the likelihood: by conditioning on the augmented (complete) tree, one avoids having to analytically integrate over all possible unobserved histories. This makes it considerably easier to extend the model—for instance adding rate heterogeneity or piecewise sampling—without having to rederive the likelihood each time. Regarding computational efficiency, the answer depends on the scale of the hidden data. In epidemiology, for example, it would be unrealistic to simulate augmented transmission trees for large-scale epidemics in a phylodynamic MCMC. For diversification models, data augmentation allowed us to analyse cetaceans at the species level, whereas our previous analytical implementation of the OBD model in RevBayes (Andreoletti et al. 2022) had to operate at the genus level due to computational constraints (though that implementation also performed full Bayesian inference on the tree and assumed homogeneous rates, so the two are not directly comparable). In summary, we recommend data augmentation when model flexibility is a priority and the augmented state space remains tractable, but analytical likelihoods when they are available and scalability is the main concern. We have added a paragraph discussing this trade-off in the Conclusion.

R2.5 – Figure 2a (truth recovery) to supplementary

> *I'm not so sure how much panel (a) adds to the narrative of the main text of the manuscript, it could be moved to the supplementary if there are space constraints.*

[TODO]

R2.6 – Figure 4 runtime discrepancy

> *Figure 4: The figure makes it seem as if an OBDD analysis of a 400-tip tree is possible in three days. However, according to Table 1, the empirical OBDD analysis took 100 days. [...] Can this difference be explained by difference in diversification and fossil sampling rates (more augmented branches in the empirical analysis)?*

The reviewer is correct that the discrepancy is driven by the number of augmented branches. Cetaceans experienced a pronounced boom-and-bust diversification history, with a large radiation followed by widespread extinction. This means that many lineages existed in the past but left no extant descendants and were not sampled as fossils, requiring the MCMC to simulate a large number of augmented branches to account for these unobserved lineages. Under the geometric Brownian motion (GBM) prior used to generate the calibration datasets, such extreme fluctuations in diversification rates are unlikely: the GBM produces relatively smooth, gradual rate variation, so the simulated trees rarely exhibit the dramatic rise and decline seen in cetaceans. As a result, the calibration datasets contain far fewer unobserved lineages and thus fewer augmented branches, leading to shorter per-iteration runtimes. In addition, reaching convergence on the cetacean dataset required more MCMC iterations, because the large number of augmented branches expands the parameter space that the chain must explore. We have clarified this point in the revised manuscript.

R2.7 – Figure 7 tick labels cut off

> *Figure 7: The tick labels on the color axes are cut off for some panels, which is a bit distracting. Please try to un-crop the tick labels.*

Done. We have increased the right margin of the tree panels so that the colorbar tick labels are no longer cropped.

R2.8 – Figure 7 linked color maps

> *If the color maps within each row would be linked, it would be easier to compare for example column 1 to the other columns, and the figure would be more pedagogic in terms of answering questions such as "what is the impact of including occurrence/phylogenetic fossils?"*

Done. The color scales are now shared within each row of Figure 7: all speciation rate panels use the same color range, and likewise for the extinction rate and net diversification panels. This makes cross-model comparisons straightforward.

R2.9 – Speciation event: rate inheritance

> *Lines 138–139: Consider saying what happens at a speciation event. I assume that the speciation and extinction rates on the child branches are inherited from the parent.*

Done. We now explicitly state that at each speciation event, both daughter lineages inherit the speciation and extinction rates of the parent lineage.

R2.10 – Time discretization δt

> *Lines 167–168: How is the size of the time discretization chosen or set?*

We have added a parenthetical in the Methods stating that δt is set to 10^{-3} Myr (i.e. 1000 years) in all our analyses.

R2.11 – Stem branch phrasing

> *Lines 274–275: The wording of the sentence makes it sound as if the stem branch, meaning for example the time of MRCA is being inferred. My understanding is that the reconstructed phylogeny is fixed and that the diversification rates are inferred. Consider re-phrasing to say "diversification rates on the stem branch can be inferred" or similar.*

Rephrased to clarify that it is the diversification rates on the stem branch that can be inferred when uncoded occurrences are available, not the topology itself.

R2.12 – Appendix: clarify "branch b"

> *Appendix A, line 680: It is not completely clear to me whether "branch b" is the "reconstructed-tree branch" or "augmented branch". [...] It might be an idea to use a bit more explicit terminology when referring to the branch.*

Clarified. "Branch b" refers to a reconstructed-tree branch. We have updated the terminology in Appendix A to explicitly state "reconstructed-tree branch" to avoid ambiguity.

R2.13 – Bibliography errors

> *There are many errors in the main text publication list, missing issue/volume/page numbers, inconsistent use of links, and more. This should be looked after more carefully.*

[TODO]

R2.14 – Custom tree plots

> *Not a criticism but I am curious: I see that in Figure 7 the diversification rates change along the branch, which is to my knowledge not possible to plot using ggtree. Did the authors implement their own tree plots from scratch?*

Yes, these tree plots are generated directly by the Tapestree.jl package, which implements custom plotting functions for augmented trees with branch-specific rate gradients.

Reviewer 3 (Dr. Peter Wagner)

We thank Dr. Wagner for his thorough review and for drawing our attention to relevant paleobiological literature. We address his major themes and specific comments below.

R3.1 – Sampling model and sampling distributions

The reviewer raises two distinct points regarding the sampling model: (1) that sampling is fundamentally binomial rather than Poisson, and (2) that a lognormal distribution of sampling rates among taxa is better supported empirically than a Gamma distribution.

Poisson vs. binomial sampling

The reviewer notes that our occurrence sampling equation is "for all intents and purposes, identical" to the TRiPS method (Starrfelt & Liow 2016), and is a variant of Huelsenbeck & Rannala's (1997) topology likelihood given stratigraphic data. He points out that sampling is fundamentally binomial (dependent on the number of available sites/collections), not Poisson.

[TODO]

Gamma vs. lognormal distribution for rate heterogeneity among taxa

The reviewer argues that a lognormal distribution of sampling rates among taxa is better supported empirically than a Gamma distribution (Wagner & Marcot 2013; Foote 2016).

We chose the Gamma distribution for the among-lineage variation in sampling rates primarily because it is conjugate to the Poisson likelihood, which allows the use of Gibbs sampling for efficient posterior inference. This conjugacy substantially speeds up the MCMC, which is critical given the already high computational cost of the OBDD model. In practice, the Gamma distribution provides a flexible unimodal shape that closely approximates the lognormal for the parameter ranges typically encountered (both are right-skewed, unimodal distributions on the positive reals), so the practical difference between the two is expected to be small. We note, as an aside, that in an ongoing project on the fossil record of Cenozoic planktonic foraminifera, a Gamma distribution was found to provide a substantially better fit to the empirical distribution of fossil sampling rates than a

lognormal. This suggests that the Gamma is not merely a computational convenience but can be a genuinely adequate model for sampling rate heterogeneity.

R3.2 – Coded vs. uncoded taxa: separate sampling distributions

The reviewer challenges our use of separate sampling rates for coded fossils vs. uncoded occurrences. His Figure 1 shows the distribution of PBDB occurrences for coded and uncoded cetacean OTUs, demonstrating that "the uncoded taxa are more apt to come from those known from fewer collections. This does not represent a separate sampling distribution: there are no taphonomic differences between these taxa and others." He argues that the distinction reflects human coding effort (well-sampled species get coded first) rather than different biological or taphonomic sampling regimes.

We appreciate the reviewer's detailed argument on this point, including his Figure 1 showing the distribution of coded and uncoded cetacean OTUs by number of collections. This critique prompted us to look more carefully at what our parametrization actually assumes, and we believe the concern can be addressed through a simple reparametrization.

In our model, for a lineage of length L in a given time period, coded fossils arise as a Poisson process with rate ψ and uncoded occurrences as an independent Poisson process with rate ω. That is, $F \sim \text{Pois}(\psi L)$ and $O \sim \text{Pois}(\omega L)$. Now define:

- a *total fossil sampling rate*: $\varphi = \psi + \omega$,
- a *coding probability*: $p = \psi/(\psi + \omega)$.

By a standard property of independent Poisson variables, the total number of finds $K = F + O$ follows:

$$K = F + O \sim \text{Pois}(\varphi L),$$

and, conditional on the total $K = k$, the number of coded fossils is Binomial:

$$F \mid K = k \;\sim\; \text{Binom}\left(k,\, \frac{\psi}{\psi + \omega}\right) = \text{Binom}(k,\, p).$$

This reparametrization is exact: the two formulations yield identical likelihoods. So our model with separate ψ and ω is strictly equivalent to a model with a single total sampling rate φ and a probability p that each find is coded.

Put differently, the total number of fossil finds in our model is governed by a single Poisson process reflecting the number of fossiliferous collections, and each find is then coded or not with probability p (a Binomial draw), compatible with the reviewer's point that sampling is fundamentally "a binomial (or multinomial) one dependent upon the number of sites/collections." Our model does *not* assume that coded and uncoded fossils come from different taphonomic or environmental sampling regimes. Rather, it assumes a single fossil discovery process and a coding probability, consistent with the idea that the distinction between coded and uncoded taxa reflects collection intensity and coding effort, not a separate sampling distribution.

Where our model goes slightly further is that ψ and ω (and hence φ and p) can vary in a piecewise manner across user-defined time periods. In other words, we do not force the coding probability p to be the same across all geological periods. In some periods, the proportion of fossils that get coded may be higher, for instance because better preservation yields more complete specimens suitable for morphological study, or because certain intervals have received more attention from the paleontological community.

We agree with the reviewer that for cetaceans specifically, the coding probability p likely does not vary much across time periods, given the extensive and relatively uniform taxonomic effort on this group. We have revised the manuscript to make clear that our parametrization does not imply separate sampling processes for coded and uncoded fossils, and we acknowledge this point in the Discussion. That said, the OBDD model is designed as a general framework for diverse clades, not only cetaceans, and we cannot assume that the coding probability is constant across time periods for all groups this model could be applied to. By not forcing p to be constant, we let the data speak to whether the coding probability varies, rather than imposing an assumption that may not hold universally. The model can recover a constant p if the data support it, but is not forced to assume one.

R3.3 – Expected sampling and taphonomic controls

The reviewer emphasizes that expected sampling can and should be assessed in advance using taphonomic controls—localities where the focal clade *could* have been found but was not (Bottjer & Jablonski 1988; Alroy et al. 2001). He provides a figure (his Figure 2) showing the number of relevant fossiliferous collections over time from the PBDB, with two levels of taphonomic control (marine tetrapods and marine vertebrates). He argues that this provides direct information about expected sampling rates, reduces the need for complex model-based estimation, and that "a clade's 'entire fossil record' is not just [...] the localities yielding [the clade], it includes the localities that *could* have yielded [fossils] but failed to do so."

[TODO]

R3.4 – Phylogenies are not the only way to infer missing taxa

The reviewer notes that other birth-death-sampling methods—such as three-timers (Alroy 2008), capture-mark-recapture (Connolly & Miller 2001), forward & reverse survivorship (Foote 2001), and TRiPS (Starrfelt & Liow 2016)—also explicitly account for unsampled lineages when estimating richness or diversification rates. He also points out an important limitation of phylogenetic approaches: they infer unsampled lineages "in one temporal direction only," meaning they can account for "Signor-Lipps lineages" (ghost lineages extending ranges backward in time) but not unsampled dead-end lineages appearing *after* sampled taxa.

[TODO]

R3.5 – Insufficient engagement with the paleobiological literature

The reviewer identifies multiple instances where our paper does not credit prior paleobiological work. Key examples include:

- Rate heterogeneity in diversification has been modeled in paleontology for over 40 years (e.g., Sepkoski 1981).
- Logistic diversification with richness-dependent origination was shown for mammals by Alroy (1996), predating similar findings by Quintero et al.
- The observation that "diversity expansions are fueled by bursts of rapid speciation, while slowdowns and declines are primarily governed by extinction rates" was first demonstrated by Gilinsky & Bambach (1987).

- The recognition that fossil sampling varies over time predates Dominici et al. (2020) by decades (Raup 1975).

[TODO]

R3.6 – Why use whales?

The reviewer asks that we explicitly explain why cetaceans are a good model group, noting that they have one of the most complete fossil records in the PBDB thanks largely to the efforts of Mark Uhen, with extensive taxonomic curation. He recommends citing Uhen et al. (2023) for the PBDB and acknowledging the top data contributors.

[TODO]

R3.7 – Terminology: "phylogenetic fossil" vs "occurrence"

The reviewer finds our terminology "awful and confusing," noting that "a fossil occurrence means that a species is known from a fossil at a fossiliferous locality. That is the way we've used the term for decades." He suggests using "coded taxa" and "uncoded taxa" instead.

We fully agree and have adopted the terms "coded fossils" and "uncoded occurrences" throughout the manuscript (see response to the Associate Editor).

R3.8 – "Incidence" synonymous with "occurrence" (p. 3, l. 52)

The reviewer notes that "incidence" is synonymous with "occurrence" in macroecology and paleobiology, and that the parallel to phylodynamics should clarify this for readers.

[TODO]

R3.9 – Rate heterogeneity in paleontology (p. 4, l. 71–73)

Regarding our statement about progress in developing diversification models with rate heterogeneity, the reviewer notes: "maybe the authors mean in the context of phylogenies: but with general data, paleontologists have been using models that demonstrate rate heterogeneity for over 40 years" (e.g., Sepkoski 1981).

[TODO]

R3.10 – Logistic diversification (p. 5, l. 96–100)

The reviewer argues that the pattern described for Quintero et al. (2024) was "originally shown for mammals 30 years ago" by Alroy (1996) and that this constitutes logistic diversification, a concept widely discussed in paleontological literature.

[TODO]

R3.11 – Slowdowns governed by extinction (p. 5, l. 102–105)

The reviewer points out that the finding attributed to Quintero et al. (2025) about diversity expansions driven by speciation bursts and declines governed by extinction "was first demonstrated nearly 40 years ago by Gilinsky & Bambach (1987)."

[TODO]

R3.12 – Single-specimen morphospecies (p. 5, l. 106–108)

Regarding our statement that "morphospecies are often only represented by a single described specimen," the reviewer notes that nearly 50% of OTUs in the whale matrices include polymorphic characters (requiring 2+ specimens). He asks that we "phrase this in a way to make it clear that single-specimen taxa are not the overwhelming majority of analyzed taxa in most phylogenetic studies."

[TODO]

R3.13 – Variable sampling rates (p. 7, l. 145–146)

Our citation of Dominici et al. (2020) for the recognition that fossil sampling varies over time is criticized: "did it really take us until 2020 to realize that?" The reviewer points to Raup (1975) as the earliest reference and notes that the PBDB was founded in 1998 specifically to address temporal sampling heterogeneities.

We have added the citation to Raup (1975) alongside Dominici et al. (2020).

R3.14 – "Complete phylogeny" (p. 8, l. 158–159)

The reviewer notes that the augmented "complete phylogeny" still omits Signor-Lipps lineages and completely unsampled non-ancestral taxa, so "it is really not complete at all."

[TODO]

R3.15 – Likelihood vs probability (p. 9, l. 185–186)

> *It is the tree and the sampling rates that have likelihoods, not the data. Thus, it should state "The probability of the whole occurrence dataset…" or "The likelihood of the tree and sampling rates".*

Corrected. We now write "the probability of the [uncoded] occurrence data Ω…"

R3.16 – 405 fossils (p. 12, l. 253–255)

The reviewer clarifies that our "405 fossils (representing distinct morphotaxa)" should read "405 fossil taxa representing distinct morphotaxa (or morphotypes)," since these almost certainly represent many more individual fossils.

Rephrased to "405 coded fossil taxa, each representing a distinct morphotaxon."

R3.17 – PBDB citation (p. 12, l. 259–260)

The reviewer states that the appropriate reference for the PBDB is Uhen et al. (2023, *PaleoBios* 40:1), which outlines how to use the PBDB. He also recommends citing the top 10 data contributors and obtaining a PBDB publication number.

We have added the citation to Uhen et al. (2023). We will acknowledge the top PBDB contributors and obtain a publication number upon acceptance.

R3.18 – "Diversification models" (p. 12, l. 264)

The reviewer argues that our eight fitted models are "not really diversification models" in the classical sense, and that true diversification models would include exponential vs. logistic vs. hierarchical diversification patterns, alternative extinction peaks, etc.

[TODO]

R3.19 – Epoch names and intervals (p. 13, l. 279–282)

The reviewer asks that epoch names be rephrased: "The Rupelian is equivalent to the Early Oligocene. However, the Aquitanian and the Messinian are only parts of the Early Miocene and Late Miocene, respectively. So, write 'Aquitanian (Early Miocene) and Messinian (Late Miocene)' as more of your readers will know when the Miocene is than when the Aquitanian or Messinian."

Done. We have rephrased to "Rupelian (Early Oligocene), Aquitanian (Early Miocene), and Messinian (Late Miocene)."

R3.20 – Occurrence rates higher "as expected" (p. 19, l. 361)

The reviewer argues that higher uncoded occurrence rates are not a biological expectation but merely a consequence of coding well-sampled species first, reinforcing that there should not be two separate sampling rates.

[TODO]

R3.21 – Rupelian date (p. 19, l. 365)

The reviewer questions our Rupelian end date of 28.1 Ma, noting that the Geologic Time Scale 2020 (Gradstein et al.) gives 33.9–27.29 Ma for the Rupelian.

Corrected to 33.9–27.3 Ma following Gradstein et al. (2020).

R3.22 – Alternative occurrences: teeth, scales, etc. (p. 23, l. 408–410)

The reviewer notes that teeth are a poor example for mammals (they are heavily used in phylogenetic analyses), and that biochemical proxies and trace fossils also have significant limitations. He suggests that the key constraint is whether taphonomically controlled sampling opportunities exist.

[TODO]

R3.23 – Phylogenetic framework not unique for unsampled lineages (p. 23, l. 419–421)

The reviewer reiterates that other BDS methods also infer unsampled lineages, and are moreover agnostic about whether unsampled lineages are older or younger than sampled taxa.

[TODO]

R3.24 – Heterogeneous rates (p. 23, l. 424)

Regarding our section heading "Heterogeneous rates give us a refined view," the reviewer notes: "paleontologists have known this for decades."

[TODO]

R3.25 – Declining net-diversification and logistic diversification (p. 24, l. 435–437)

The reviewer argues that our observed pattern of declining net-diversification rates is "almost certainly picking up logistic diversification" and recommends engaging with the literature on diversity-dependent models.

[TODO]

R3.26 – Offer to discuss

The reviewer offers to discuss the use of stratigraphic databases for refining PBDB occurrences and notes that while the review may come across as negative, the underlying goal is to encourage engagement between phylogenetic and paleobiological approaches.

[TODO]